Building Characteristics and Energy Use in New York City

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Academic Project Report

Abstract

This study examines how building characteristics relate to energy use in New York City. Using the city's Local Law 84 energy benchmarking disclosures, it builds a data-cleaning workflow, a normalized MySQL database, and three regression models (linear, log-linear, and polynomial), and uses partial R^2 to measure the unique contribution of each predictor. In a cleaned sample of 50,635 benchmarking records from 22,684 buildings, gross floor area is the most influential factor for total site energy use and explains nearly one-third of its variance on its own, while building age and grid electricity use have smaller but significant effects. The polynomial model fits best. It shows that energy use rises with floor area at a decreasing rate and follows an inverted-U pattern with building age, peaking for buildings about 80 years old.

1. Introduction

Buildings account for a large share of energy use in cities, but it is not always clear which building characteristics contribute the most. New York City's Local Law 84 (LL84) requires owners of large buildings to report their annual energy and water use, and the city publishes these benchmarking data. The disclosures have been used to predict city-scale building energy use [1], to grade buildings on energy performance [2], and to estimate the energy savings attributable to benchmarking policy [3].

The raw disclosures contain missing values, inconsistencies, and extreme outliers. This study applies systematic cleaning, stores the data in a normalized relational database, and fits three regression models (linear, log-linear, and polynomial) that relate gross floor area, building age, and grid electricity use to weather-normalized site energy use. Partial R^2 measures the unique contribution of each variable. The goal is to identify the factors that matter most for building energy use and to provide evidence for energy-efficiency planning in cities.

2. Data and Methods

2.1 Data

The data come from the LL84 energy and water data disclosures published by the NYC Department of Buildings on NYC Open Data [4]. Each record is one benchmarking submission for a property and reports, among other fields, its weather-normalized site energy use, grid electricity purchases, total gross floor area (GFA), year built, property types, and ENERGY STAR score. Weather normalization estimates the energy a property would have used under typical weather, so that years with unusually hot or cold weather are comparable [5]. The extract used here has 24 fields.

2.2 Database Construction

The cleaned records are stored in a normalized MySQL database. A Building table holds the identity and location of each property. Because a property can appear in several benchmarking records, a Benchmark_Record table links each record to its building, and structural attributes (Housing_Age, Gross_Floor_Area) and operational metrics (Electricity_Use, Total_Energy_Use, Energy_Usage_Intensity) are stored in separate tables keyed by record. A lookup table holds the national median site EUI of each property type. The modeling variables are extracted with a join across these tables.

2.3 Data Cleaning

The cleaning steps were:

- Remove records with a missing value in any modeling variable, including text placeholders such as "Not Available", "NA", "N/A", and "--".
- Keep records with a year built between 1800 and 2022.
- Remove outliers in grid electricity use, weather-normalized site energy use, and total GFA with an iterative 3σ rule: points more than three standard deviations from the mean are dropped, and the step is repeated until no point is removed.

The cleaned dataset has 50,635 records covering 22,684 buildings. Because the 3σ rule trims the long right tails of the distributions, the largest buildings are excluded: total GFA in the cleaned sample ranges from 1,289 to 142,640 ft².

2.4 Model Variables

Four variables are used, all standardized to z-scores:

- *Y*: weather-normalized site energy use (kBtu)
- *GFA*: total gross floor area (ft²)
- *Age*: 2022 minus year built
- *Electricity*: grid electricity purchases (kBtu)

2.5 Model Specifications

Linear model. The baseline specification:

$$Y = \beta_0 + \beta_1 GFA + \beta_2 Age + \beta_3 Electricity + \varepsilon$$

Log-linear model. Tests whether energy use scales with the logarithm of building size:

$$Y = \beta_0 + \beta_1 log(GFA) + \beta_2 Age + \beta_3 Electricity + \varepsilon$$

Polynomial model. Adds squared terms to capture nonlinear effects of size and age:

$$Y = \beta_0 + \beta_1 GFA + \beta_2 GFA^2 + \beta_3 Age + \beta_4 Age^2 + \beta_5 Electricity + \varepsilon$$

All models are estimated by ordinary least squares. In the polynomial model, the age at which predicted energy use peaks follows from setting the derivative with respect to age to zero, after converting the standardized coefficients back to years.

2.6 Partial R²

Partial R² measures the share of the variance left unexplained by a reduced model that a predictor (or group of predictors) explains when it is added back:

$$R^2_{partial} = \frac{R^2_{full} - R^2_{reduced}}{1 - R^2_{reduced}}$$

For the polynomial model, predictors are grouped as {GFA, GFA²}, {Age, Age²}, and {Electricity}.

3. Results

3.1 Sample Description

Table 1 summarizes the modeling variables. The typical record describes a building of about 52,000 ft² that uses about 3.9 million kBtu of site energy a year. The stock is old: the median building is 91 years old, and 47% of records describe buildings built before 1931. Multifamily housing dominates the sample (Table 2), followed by K-12 schools and offices. Floor area and grid electricity are correlated (r = 0.56), so part of their explanatory power is shared.

Table 1. Descriptive statistics of the modeling variables (N = 50,635)

Variable	Mean	SD	Median	Min	Max
Site energy use (kBtu)	4,263,875	2,391,762	3,938,216	0	11,862,819
Total GFA (ft²)	59,047	28,153	51,840	1,289	142,640
Age (years)	74.7	34.2	91	0	222
Grid electricity (kBtu)	1,065,351	699,806	860,946	0	3,617,806

Table 2. Records by largest property use type

Property type	Records	Share
Multifamily housing	40,333	79.7%
K-12 school	3,405	6.7%
Office	3,026	6.0%
Non-refrigerated warehouse	925	1.8%
Hotel	842	1.7%
Other (16 types)	2,104	4.2%

3.2 Model Performance

Table 3 compares the three models. The polynomial model has the highest R^2 and adjusted R^2 and the lowest AIC and BIC. The log-linear model fits slightly worse than the linear baseline.

Table 3. Model performance

Model	R^2	Adj. R^2	AIC	BIC
Linear	0.477	0.477	110,870	110,905
Log-linear	0.472	0.472	111,400	111,435
Polynomial	0.502	0.502	108,420	108,473

3.3 Polynomial Model Coefficients

Table 4 lists the standardized coefficients of the polynomial model. All terms are significant at $p < 0.001$. The linear terms of GFA and age are positive and their squared terms negative, so both effects flatten and, for age, reverse. The implied peak of predicted energy use is at a building age of 80 years, that is, for buildings built around 1942 (Figure 1); 60% of records describe buildings older than that. For floor area, the implied peak (about 207,000 ft²) lies beyond the largest building in the sample, so within the sample energy use keeps rising with size, but at a decreasing rate.

Table 4. Standardized coefficients of the polynomial model

Term	Coefficient	Std. error	t
GFA	0.848	0.015	56.8
GFA²	−0.301	0.015	−20.5
Age	0.711	0.013	56.4
Age²	−0.560	0.012	−44.9
Electricity	0.193	0.004	48.1

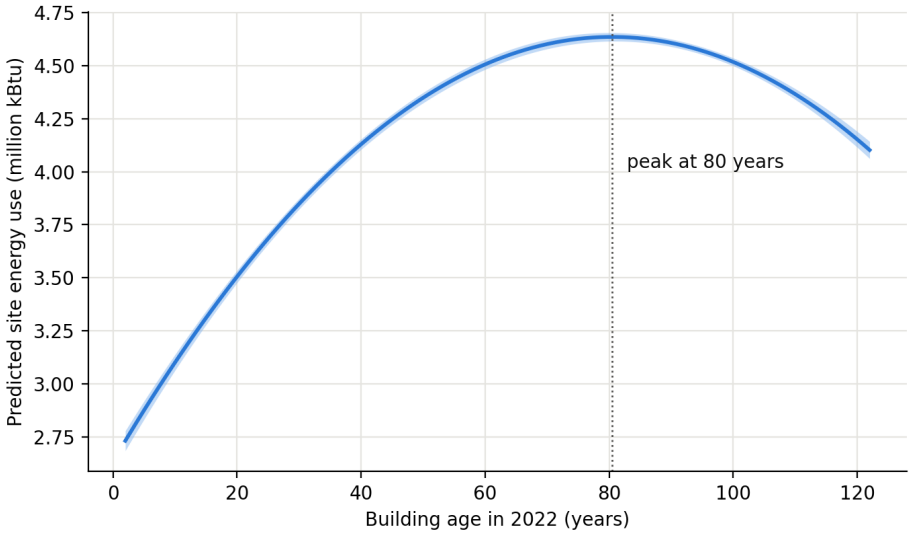


Figure 1. Predicted site energy use by building age, polynomial model (other predictors at their means; shaded band: 95% confidence interval)

3.4 Contribution of Each Predictor

Table 5 gives the partial R^2 of each predictor. Floor area explains about 32% of the variance left by the other predictors in the linear model and 29% in the polynomial model. Age matters more once its nonlinear shape is allowed (0.046 to 0.083), and the contribution of electricity also rises slightly (0.028 to 0.044).

Table 5. Partial R^2 by predictor (grouped for the polynomial model)

Predictor	Linear model	Polynomial model
Total GFA (with GFA^2)	0.3178	0.2871
Age (with Age^2)	0.0459	0.0827
Electricity	0.0282	0.0437

Figure 2 plots the residuals of the polynomial model against its fitted values. The spread of the residuals changes with the fitted value, so the errors are heteroscedastic.

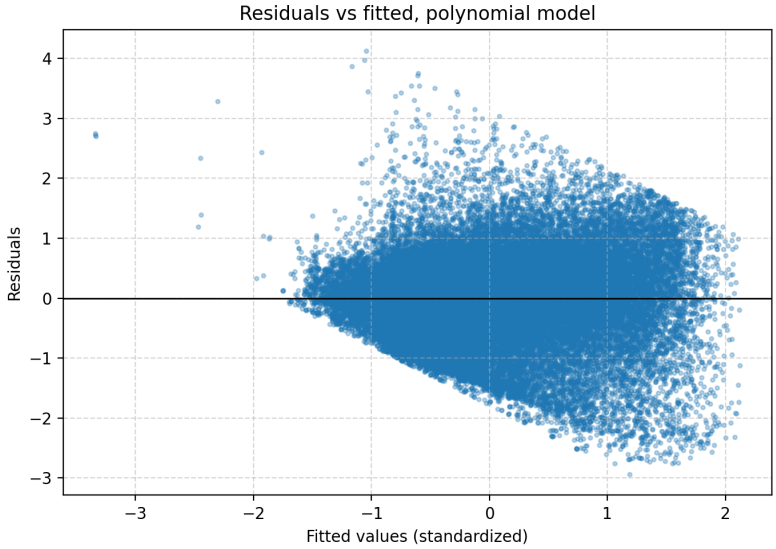


Figure 2. Residuals versus fitted values, polynomial model

4. Discussion

4.1 Building size dominates

In every specification, total GFA is the strongest predictor of total energy use: its partial R^2 shows that building size alone explains nearly one-third of the variance. This matches basic energy-modeling intuition, since larger buildings have more conditioned space, more equipment, and larger HVAC loads. The negative GFA^2 term means that each additional square foot adds less energy as buildings get larger.

4.2 An inverted-U effect of building age

Holding size and electricity use constant, predicted energy use rises with age up to about 80 years and then falls. Newer buildings are the most efficient, which is consistent with more recent construction standards. Mid-aged buildings use the most energy, possibly because of aging equipment and deferred maintenance. The oldest buildings use less again, which may reflect retrofits or survivorship, but the data contain no retrofit information to test this. Because the median building in the sample is 91 years

old, most of the stock lies on the declining side of the curve. A linear model cannot represent this pattern, which is why the effect of age nearly doubles in the polynomial model.

4.3 Electricity use plays a secondary role

Grid electricity use is significant in all models, but its partial R^2 is much smaller than those of GFA and age. It captures operational variation rather than the structural drivers of site energy use. Its contribution is also understated by its correlation with floor area, since the variance the two share is credited to neither in a partial R^2.

4.4 Choice of model

The linear model is simple and easy to interpret. The log-linear model fits slightly worse, which suggests that energy use does not scale with the logarithm of floor area. The polynomial model gives the best fit and the best information criteria and captures the nonlinear effects described above, so it is the preferred specification.

5. Limitations

- **Total energy rather than intensity.** The dependent variable is total site energy use, which grows with building size by construction. Part of the dominance of floor area is therefore mechanical. Modeling energy use intensity (EUI) instead would test whether larger buildings are more or less efficient per square foot.
- **Repeated records.** The 50,635 records come from 22,684 buildings, so observations are not independent and the reported standard errors are too small. Standard errors clustered by building would be more appropriate; the large t-statistics suggest that the main conclusions would hold.
- **Electricity as a predictor.** Grid electricity is itself part of site energy use, so its coefficient describes an accounting relationship rather than a cause. In 471 records, grid electricity exceeds the weather-normalized site energy total, which points to reporting or normalization inconsistencies.
- **Heteroscedastic errors.** The residual spread changes with the fitted value (Figure 2). This does not bias the coefficients but affects the standard errors; robust standard errors would address it.
- **Sample coverage.** Four in five records are multifamily buildings, and the 3σ trimming removed the largest buildings. The results describe mid-sized, mostly residential buildings and may not transfer to large commercial properties.
- **Associations, not causes.** The models are cross-sectional. They omit factors such as heating fuel, occupancy, and equipment, and the explanations offered for the age pattern are hypotheses.

6. Conclusion

Gross floor area is the strongest predictor of building energy use in New York City, with energy use rising at a decreasing rate as buildings get larger. Building age has a clear inverted-U effect that peaks at about 80 years, while grid electricity use plays a smaller role. The data pipeline and modeling approach provide a simple, reproducible framework for analyzing large building datasets. Future work could model energy

use intensity, account for repeated records with clustered or panel methods, add variables such as heating fuel and property type, and apply spatial and machine-learning methods.

Data and Code

The cleaning, database, and regression code, the cleaned dataset, and all tables and figures in this report are available at https://github.com/JiaweiQi3/nyc-building-energy-drivers.

References

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